Notes: Begenau and Palazzo (JFE, 2021)

Firm Selection and Corporate Cash Holdings

Contents

1 Big picture

- **Question.** Why did cash holdings of U.S. public firms rise so much after the late 1970s? Is the answer mostly that incumbent firms changed their behavior, or that the *composition* of public firms changed because different kinds of firms started going public?
- **Setting.** The paper studies U.S. public firms, mainly from Compustat, and documents the evolution of cash-to-asset ratios from 1958 to 2013, with the main structural analysis focused on 1979–2003.
- **Core challenge.** A rise in average cash can come from two very different sources: a *within-firm effect* (existing firms accumulate more cash over time) or a *selection effect* (new entrants have systematically higher cash balances than older incumbents). Reduced-form evidence alone cannot quantify how important each force is in the secular trend.
- **Main stylized facts.** The paper emphasizes three facts. First, the rise in average cash is driven by firms entering public markets with progressively higher cash balances. Second, once public, firms on average *decrease* their cash-to-asset ratios over time. Third, the overall rise in average cash is mostly a selection effect, and that selection effect is driven overwhelmingly by **R&D-intensive** entrants.
- **Main mechanism.** The authors embed entry into a standard dynamic corporate finance model. Firms face decreasing returns to scale, persistent but mean-reverting productivity, capital adjustment costs, costly external equity, and a lower return on internal cash. Entrants with low initial productivity signals expect fast future growth and therefore choose to go public with high cash balances as a buffer against future financing needs.
- **Main quantitative result.** The estimated model generates an increase in the aggregate cash-to-asset ratio of about 78% versus 160% in the data over 1979–2003, and an increase in sales-growth volatility of about 44% versus 88% in the data. Selection alone explains the majority of the model-generated rise in cash, while higher cash-flow volatility is critical for the rise in sales-growth volatility.
- **Key economic insight.** The rise in average cash is not mainly a story of mature firms accumulating cash. It is mainly a story of *smaller, riskier, less initially profitable, high-growth-potential firms* entering public markets with much larger cash buffers than earlier cohorts.
- **Why this paper matters.** It gives a structural selection-based explanation for one of the central trends in corporate finance—the secular increase in corporate cash—and shows that composition effects can be quantitatively large when studying firm behavior in public-firm data.

2 Data and measurement (key objects)

2.1 Observed public-firm panel

- **Main source.** The empirical analysis is based on Compustat data for U.S. publicly listed firms.
- **Sample.** The paper excludes financial firms (SIC 6000–6999) and utilities (SIC 4000–4999). In the time-trend regressions, firms must have positive assets and sales and at least five years of observations. In the structural estimation, firms are grouped into balanced ten-year panels after entry.

- **Entry.** For the stylized-fact analysis, firms are organized into five-year entry cohorts beginning in 1959. In the paper's main public-firm sample, entry is tied to a firm's first appearance with a year-end stock price in Compustat, and robustness checks use IPO dates.
- **Core observed variables.** The paper uses the cash-to-asset ratio, sales growth, the investment-to-asset ratio, and equity issuance scaled by assets.
- **Key aggregate fact.** The average cash-to-asset ratio of U.S. public firms rises from about 8% in 1979 to about 25% in 2013.

2.2 Cohorts, sectors, and ages

- **Cohorts.** The descriptive analysis groups firms into eleven five-year cohorts from 1959–1963 through 2009–2013. The structural model uses a baseline cohort of 1974–1978 and then separately estimates cohorts 1979–1983, 1984–1988, 1989–1993, 1994–1998, and 1999–2003.
- **R&D-intensive sector.** A firm is classified as R&D-intensive if it belongs to a three-digit SIC industry whose average R&D expenditure is at least 2% of assets over the sample period.
- **Two-sector structure.** The model collapses the data into two sectors:
  - **Industry 0:** non-R&D-intensive firms,
  - **Industry 1:** R&D-intensive firms.
- **Age bins in reduced-form evidence.** Young firms have been public at most 5 years, mature firms more than 5 but less than 16 years, and old firms at least 16 years.

2.3 Observed decomposition objects

- Let N_t^I be the number of incumbents, N_t^E entrants, and N_{t-1}^X exitors. Let CH_t^I, CH_t^E, and CH_{t-1}^X denote the average cash-to-asset ratio of incumbents, entrants, and exitors.
- The change in average cash holdings is decomposed into a within-firm term and a selection term. This decomposition is central because it measures how much of the secular rise in average cash comes from incumbents versus new firms.
- The selection term is further split into R&D-intensive and non-R&D-intensive entrants, allowing the paper to isolate which type of entrant drives the trend.

2.4 Moments used for structural estimation

- **Entry moments.** Average cash at entry, average sales growth at entry, volatility of sales growth at entry, average investment rate at entry.
- **Lifecycle moments.** Change in cash over the first ten years after entry, change in sales growth over the first ten years, change in investment over the first ten years.
- **Volatility and persistence moments.** Volatility and autocorrelation of cash holdings, sales growth, and investment.
- **Financing moment.** Average equity issuance size for mature firms.

- **Why these moments matter.** The model is disciplined not only by average levels but also by lifecycle dynamics and volatility, which helps separate selection, financing, and productivity mechanisms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Reduced-form cohort regression: the entry-margin fact

The paper first estimates the role of entry cohorts in explaining the rise in average cash holdings using

$$CH_{i,t} = \alpha + \beta t + \sum_{k \in \{K\}} \gamma_k I_{i \in k} + \varepsilon_{i,t}. \tag{1}$$

- **Interpretation.** $CH_{i,t}$ is the cash-to-asset ratio, β is the common time trend, and γ_k are cohort fixed effects.
- **Main message.** In pooled OLS, the estimated time trend is large and positive. Once cohort fixed effects are added, the time trend becomes essentially zero. So the secular rise in average cash is not primarily a within-firm trend: it is a cohort-composition effect.
- **Sector split.** The cohort fixed effects rise strongly for R&D-intensive firms but barely move for non-R&D-intensive firms, showing that the entry-margin story is concentrated in the R&D sector.

3.2 Reduced-form within-firm trend

To study whether incumbents themselves raised cash over time, the paper estimates firm-specific trends using

$$CH_{i,t} = \alpha_i + \beta_i t + \varepsilon_{i,t}, \tag{2}$$

where time is normalized to zero in the first year the firm appears in the sample.

- **Interpretation.** This specification strips out the initial cash level at entry for each firm and asks whether the firm's own cash ratio rises or falls afterward.
- **Main result.** The average within-firm trend is negative: incumbents deplete cash over time. This is especially true for R&D-intensive firms and for younger firms.
- **Economic meaning.** A firm that enters public markets with high cash is typically using that cash buffer while scaling up after IPO rather than continually accumulating more cash year after year.

3.3 Accounting decomposition of the change in average cash

The change in the cross-sectional average cash-to-asset ratio can be decomposed as

$$\Delta CH_t = \left(\frac{N_t^I}{N_t} CH_t^I - \frac{N_{t-1}^I}{N_{t-1}} CH_{t-1}^I \right) + \left(\frac{N_t^E}{N_t} CH_t^E - \frac{N_{t-1}^X}{N_{t-1}} CH_{t-1}^X \right). \tag{3}$$

A further split by entrant type gives

$$\Delta CH_t = \left(\frac{N_t^I}{N_t} CH_t^I - \frac{N_{t-1}^I}{N_{t-1}} CH_{t-1}^I \right) + \sum_{i \in \{\text{R\&D}, \text{nonR\&D}\}} \left(\frac{N_t^{E_i}}{N_t} CH_t^{E_i} - \frac{N_{t-1}^{X_i}}{N_{t-1}} CH_{t-1}^{X_i} \right). \tag{4}$$

- **Interpretation.** The first bracket is the within-incumbent contribution. The second bracket is the selection contribution from entrants net of exitors.
- **What the decomposition reveals.** The selection effect from R&D-intensive entrants contributes more than 200% of the secular rise in cash, the selection effect from non-R&D entrants contributes less than 40%, and the within-incumbent component contributes −117%.
- **Economic meaning.** Average cash rises *despite* incumbents running cash down, because high-cash new entrants progressively dominate the public-firm sample.

3.4 Incumbent firm's problem

The model is a standard neoclassical investment problem with cash choice and costly external finance.

Technology. Firms produce with decreasing returns to scale:

$$y_t = p e^{z_t} k_t^{\alpha}, \qquad z_{t+1} = \rho z_t + \sigma \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0,1). \tag{5}$$

Capital evolves according to

$$k_{t+1} = (1-\delta) k_t + x_{t+1}, \tag{6}$$

with adjustment cost

$$\phi(k_{t+1}, k_t) = \eta \left(\frac{k_{t+1} - (1-\delta) k_t}{k_t} \right)^2 k_t. \tag{7}$$

- **Interpretation.** Firms face persistent productivity shocks, and capital is costly to adjust. This gives a dynamic reason for carrying liquid assets.
- **Role of decreasing returns.** Firms that start small have high marginal product of capital and therefore strong growth incentives.

Financing. Cash carried internally earns a lower return than the risk-free rate, summarized by a wedge $\nu \in (0,1)$. External equity issuance is costly. If dividends are negative, the firm is issuing equity and pays

$$H(d_t) = -f_e - \kappa |d_t|. \tag{8}$$

- **Interpretation.** Holding cash is not free, but external equity is also costly. The optimal cash policy trades off carry costs against future financing costs.

Value function. Let $V_t = V(k_t, c_t, z_t)$. The incumbent solves

$$V_t = \max_{c_{t+1} \geq 0,\, x_{t+1}} \left\{ d_t + H(d_t) \mathbf{1}_{\{d_t \leq 0\}} + \frac{1-\lambda}{R} E_t[V_{t+1}] + \frac{\lambda}{R} E_t[w_{t+1} + (1-\delta) k_{t+1}] \right\}, \tag{9}$$

subject to

$$d_t = w_t - \frac{c_{t+1}}{R} - x_{t+1} - \phi(k_{t+1}, k_t), \tag{10}$$

$$k_{t+1} = (1-\delta)k_t + x_{t+1}, \tag{11}$$

$$w_{t+1} = pe^{z_{t+1}} k_{t+1}^{\alpha} + c_{t+1}. \tag{12}$$

- **Interpretation.** Net worth can be distributed as dividends, invested into capital, or stored as cash. Firms exit exogenously with probability λ.
- **Why cash matters.** Cash is a buffer stock that helps the firm avoid costly equity issuance when productivity is currently low but future investment needs are high.

3.5 Entry and the selection mechanism

Exiting firms are replaced by an equal mass of heterogeneous entrants. Each entrant receives a productivity signal q drawn from a Pareto distribution on $[\underline{q}, \infty)$ with shape parameter ξ. The entrant chooses initial capital and cash according to

$$V^E(q_t) = \max_{c_{t+1}, k_{t+1}} \left\{ -k_{t+1} - \frac{c_{t+1}}{R} + \frac{1}{R} E\big[V(k_{t+1}, c_{t+1}, z_{t+1}) \mid q_t\big] \right\}, \tag{13}$$

with

$$z_{t+1} = \rho \log q_t + \sigma \varepsilon_{t+1}. \tag{14}$$

- **Interpretation.** Entry is not an equilibrium IPO decision model. Instead, it is a tractable mechanism that lets the distribution of entrant types change over time.
- **Economic intuition.** A lower signal q implies lower initial productivity. With mean reversion, that entrant expects strong growth and therefore wants to enter with high cash balances.
- **Selection mechanism.** If later cohorts draw lower average signals, then later public entrants hold more cash at IPO even if the incumbent optimization problem is unchanged.

3.6 Model implications for cash holdings

- **Incumbents.** Large, high-productivity firms with large installed capital have low precautionary savings demand because they expect high cash flows and low growth needs. Small firms have the opposite pattern.
- **Entrants.** High-signal entrants invest heavily and carry little cash. Low-signal entrants choose lower initial capital and higher cash-to-asset ratios.
- **Key joint implication.** A downward shift in entrant productivity signals naturally generates two facts simultaneously:
  - a positive cross-cohort trend in cash at entry,
  - a negative within-firm trend in cash after entry.

3.7 Structural estimation

The model is estimated by simulated method of moments (SMM). A generic representation is

$$\hat{\theta} = \arg\min_{\theta} \Big(m^{\text{data}} - m^{\text{sim}}(\theta)\Big)' W \Big(m^{\text{data}} - m^{\text{sim}}(\theta)\Big). \tag{15}$$

- **Estimated parameters.** For the baseline cohort, the paper estimates eight parameters: α, η, ρ, σ, ν, f_e, $\underline{q}$, and ξ.
- **Calibrated parameters.** The depreciation rate is set to $\delta = 0.15$, the interest rate to $r = 0.04$, the proportional equity issuance cost to $\kappa = 0.07$, the scale parameter to $p = 0.07$, and the exogenous exit rate to $\lambda = 0.08$.
- **Industry-cohort estimation.** After the baseline cohort, the paper estimates the model separately by industry and cohort. The Pareto shape parameter ξ is fixed at its baseline estimate because it is weakly identified.

4 Identification and moments (what makes the estimates credible)

4.1 What the reduced-form evidence identifies

- The regression with cohort fixed effects shows that the aggregate trend in cash disappears once entry cohorts are controlled for. This is direct evidence that the rise in average cash is a composition effect rather than a common time effect.
- The firm-by-firm regressions show that the average within-firm trend is negative. This rules out the simple interpretation that mature public firms steadily raised cash over time.
- The decomposition into within and selection terms quantifies the magnitude of the selection mechanism directly in the data before any structure is imposed.

4.2 What identifies the structural parameters

- **Persistence and volatility of productivity** (ρ, σ)**.** Identified by the autocorrelation and volatility of sales growth.
- **Returns to scale** (α)**.** Identified by the change in sales growth over the lifecycle, because decreasing returns make firms' growth rates fall as they scale up.
- **Adjustment costs** (η)**.** Identified by volatility of the investment-to-asset ratio: high adjustment costs smooth investment.
- **Financing frictions** (ν, f_e)**.** Identified by average cash at entry and by the size of equity issuance among mature firms.
- **Entry distribution** $(\underline{q}, \xi)$**.** Identified by average and volatility of sales growth at entry, because entrants' initial productivity determines both their scale and their growth prospects.

4.3 Why the industry split is essential

- The descriptive evidence shows that the secular rise in cash is almost entirely a phenomenon of R&D-intensive firms.
- Estimating one pooled model would miss the fact that later R&D-intensive entrants are smaller, riskier, and more cash-heavy than earlier cohorts, while non-R&D cohorts change much less.
- The sector split is therefore not cosmetic. It is the main device that links the reduced-form facts to the structural mechanism.

4.4 What is not fully captured

- The model makes firms deplete cash too quickly relative to the data. So the negative within-firm effect is too strong quantitatively, even if it is correct qualitatively.
- The Pareto shape parameter ξ is weakly identified. The paper therefore fixes it in the industry-cohort estimation.
- The model is not a full IPO decision model and does not endogenize the mass of entrants. It is instead a tractable device for studying how entrant composition affects aggregate cash holdings.

4.5 Relation to the literature

- Relative to **Bates et al.** and related reduced-form papers, this paper isolates the selection mechanism in a structural framework.
- Relative to **Riddick and Whited**, it adds heterogeneous entry and allows the distribution of entrants to move over time.
- Relative to the broader cash-holdings literature, the paper shows that sample composition can be quantitatively central, especially because of the growing role of R&D-intensive firms among public firms.

5 Tables and figures

5.1 Figure 1 and Table 1: the aggregate rise in cash is really a cohort effect

Read:

- Figure 1 shows the headline fact: average cash-to-assets rises from about 8% in 1979 to about 25% by 2013.
- Table 1 shows that the pooled time trend is about 0.415 percentage points per year, but once cohort fixed effects are included, the trend becomes essentially zero.
- This is the first major empirical punchline of the paper: the secular rise in cash is not a common trend across incumbent firms; it is driven by who enters the public-firm sample.

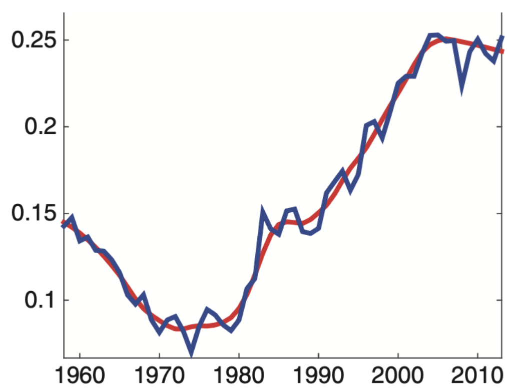

Fig. 1. Average cash-to-assets ratio of US-listed firms. This figure reports the average cash-to-assets ratio of US public companies over the period 1958-2013.

Table 1
Estimating the entry cohort effect.

	Pooled OLS I	All firms II	Non-R&D-intensive III	R&D-intensive IV
t	0.415***	−0.000	−0.033*	0.017
	(0.019)	(0.019)	(0.019)	(0.032)
1959-1963		−1.521*	−0.931	−1.811
		(0.916)	(1.103)	(1.499)
1964-1968		−1.335	−0.322	−2.274
		(0.921)	(1.148)	(1.407)
1969-1973		−0.932	−1.130	0.705
		(0.840)	(0.952)	(1.473)
1979-1983		6.288***	2.445**	8.547***
		(0.980)	(1.113)	(1.519)
1984-1988		7.970***	1.263	13.711***
		(1.000)	(0.993)	(1.602)
1989-1993		10.739***	0.527	19.177***
		(1.053)	(1.004)	(1.608)
1994-1998		13.513***	2.183**	22.126***
		(1.043)	(1.074)	(1.560)
1999-2003		23.835***	7.186***	29.465***
		(1.299)	(1.562)	(1.705)
2004-2008		16.569***	2.678**	28.328***
		(1.513)	(1.326)	(2.222)
2008-2013		14.255***	3.667	22.919***
		(3.805)	(2.606)	(5.731)
Constant	11.678***	11.349***	10.270***	13.022***
	(0.342)	(0.719)	(0.817)	(1.210)
Observations	76,872	76,872	38,553	38,319
R-squared	0.031	0.108	0.016	0.163

We estimate the following baseline linear equation:

$$CH_{i,t} = \alpha + \beta t + \sum_{k \in \{K\}} \gamma_k \times I_{i \in k} + \varepsilon_{i,t}.$$

The dependent variable is the cash-to-assets ratio defined as Compustat item CHE divided by Compustat item AT and expressed in percentage terms. The sample includes Compustat firm-year observations from 1979-2013 with at least five years of observations and positive values for assets and sales, excluding financial firms (SIC codes 6000 to 6999) and utilities (SIC codes 4000 to 4999). Differently from the regression in Table 2, we exclude firms that entered Compustat before 1959. In column 1, we run pooled OLS regressions, and we normalize the year 1979 to zero. In column 2, we report the results with cohort fixed effects. In column 3, we report the results with cohort fixed effects only for the R&D-intensive sector, whereas in column 4, we report the results with cohort fixed effects only for the non-R&D-intensive sector. We report standard errors that are clustered at the firm level. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

5.2 Figure 2 and Table 2: entrants come in with more cash, incumbents run it down

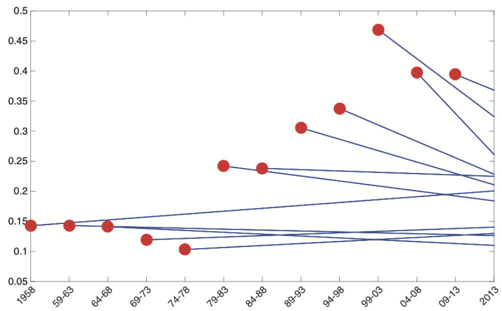

Fig. 2. Average cash holdings at entry (1959-2013). The figure reports the evolution of the cash-to-assets ratio around IPO for US public companies for 11 five-year cohorts over the period 1959-2013. The red dot denotes the average cash ratio at entry for each cohort. The first observation denotes the average cash holdings of incumbent firms in 1958. The straight line connects the initial average cash holdings to the average holding in 2013 for each cohort. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2
Estimating the time trend.

	Pooled OLS		Firm-by-firm							
	All I	All II	All III	All IV	Young V	Young VI	Mature VII	Mature VIII	Old IX	Old X
Trend	0.418*** 0.007	0.055*** 0.006	−0.497*** 0.034	−0.267*** 0.048	−0.461*** 0.037	−0.239*** 0.053	−0.226*** 0.026	−0.115*** 0.037	0.005 0.239	0.001 0.033
Trend X R&D		0.605*** 0.013		−0.463*** 0.068		−0.447*** 0.075		−0.224*** 0.052		0.007 0.048
R&D dummy		4.579*** 0.253		20.711*** 0.587		20.760*** 0.642		0.180*** 0.000		11.100*** 0.910
Constant	10.661*** 0.123	9.371*** 0.122	21.920*** 0.325	11.640*** 0.413	21.581*** 0.350	11.276*** 0.453	20.234*** 0.387	11.446*** 0.505	14.648*** 0.474	9.510*** 0.619
Observations	85,947	85,947	(5,496; 16)		(5,496; 5)		(3,614; 9)		(1607; 13)	
Adjusted R^2	0.035	0.185	0.295		0.418		0.312		0.291	

We estimate the following baseline linear equation:

$$CH_{i,t} = \alpha + \beta t + \varepsilon_{i,t}.$$

The dependent variable is the cash-to-assets ratio defined as Compustat item CHE divided by Compustat item AT and expressed in percentage terms. The sample includes Compustat firm-year observations from 1979-2013 with at least five years of observations and positive values for assets and sales, excluding financial firms (SIC codes 6000 to 6999) and utilities (SIC codes 4000 to 4999). We also sort firms into R&D- versus non-R&D-intensive, as discussed in Section 2.3. In columns 1 and 2, we run pooled OLS regressions and we normalize the year 1979 to zero. In columns 3-10, we run a linear regression for each firm in our sample and set the time variable equal to zero the first year the firm appears in the sample. Young firms are firms that have been public for at most 5 years, mature firms are firms that have been public for more than 5 years but less than 16 years, and old firms are firms that have been public for at least 16 years. For the firm-by-firm regressions, we report the number of individual firms in the sample together with the average number of observations for each firm. The reported R^2 for the firm-by-firm regressions is the average R^2 across all the regressions. We report robust standard errors. * denotes significance at the 10% level, ** denotes significance at the 5% level, and *** denotes significance at the 1% level.

Read:

- Figure 2 shows that entry cohorts after the late 1970s go public with progressively higher cash ratios.
- Table 2 shows that the average pooled trend is positive, but the *within-firm* trend is negative at about −0.497 percentage points per year on average.
- The same table also shows that the R&D sector has a much steeper positive pooled trend than the non-R&D sector (roughly 66 basis points versus 6 basis points per year), while R&D firms also deplete cash faster after entry.
- The combination of Figure 2 and Table 2 creates the core identification logic for the paper: entrants are driving up the average while incumbents are moving the other way.

5.3 Figures 3, 4, and 5: R&D-intensive entrants drive the selection effect

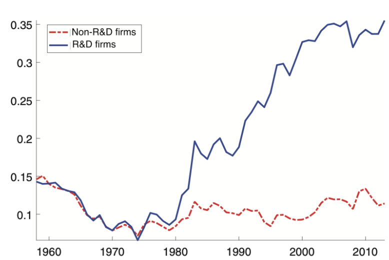

Fig. 3. Average cash-to-assets ratio of US listed firms by sector. This figure reports the average cash-to-assets ratio of R&D-intensive and non-R&D-intensive firms over the period 1958-2013. An R&D-intensive firm belongs to an industry (three-level digit SIC code) whose average R&D investment amounts to at least 2% of assets over the sample period.

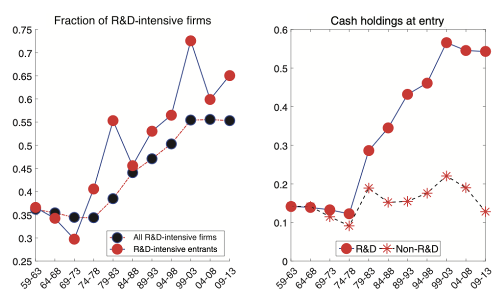

Fig. 4. Industry composition of US public firms (1959-2013). The left panel of this figure presents the share of R&D-intensive firms in Compustat (in darker color with a dashed line) and the share of R&D-intensive entrants (in red with a solid line). The right panel shows the average cash-to-assets ratio at entry of R&D-intensive and non-R&D-intensive firms. An R&D-intensive firm belongs to an industry (three-level digit SIC code) whose average R&D investment amounts to at least 2% of assets over the sample period. We group firms into cohorts of five years starting from 1959. We define as entrant a firms that reports a year-end value of the stock price for the first time (item *PRCC_C*). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Read:

- Figure 3 shows that before 1979, cash ratios of R&D-intensive and non-R&D firms were similar, but after 1979 only the R&D group experiences a large increase.
- Figure 4 shows two things at once: the share of R&D-intensive firms in Compustat rises above 55% by 2013, and the share of R&D-intensive entrants rises from about 35% to about 65%.

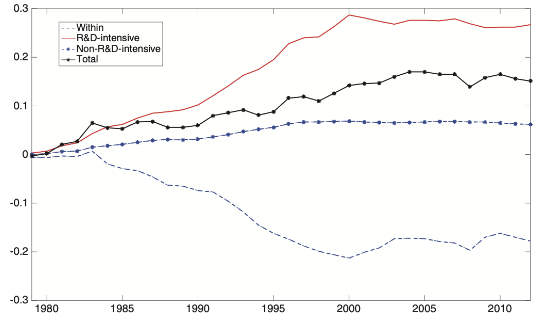

Fig. 5. Cash change decomposition. This figure reports the cumulative change in average cash holdings over the sample period together with its three components: the cumulative change due to incumbents (labeled "within"), the cumulative change due to R&D-intensive entrants, and the cumulative change due to non-R&D-intensive entrants.

- The right panel of Figure 4 shows that R&D-intensive cohorts also go public with progressively higher cash at entry, unlike non-R&D entrants.
- Figure 5 turns this into an accounting statement: the selection effect from R&D-intensive entrants contributes more than 200% of the secular rise in cash, the non-R&D selection effect contributes less than 40%, and the within-incumbent component contributes −117%.

5.4 Figure 6: the entry policy function

Read:

- Figure 6 plots optimal entry cash and capital as a function of the productivity signal.
- The logic is clean: lower-signal entrants choose lower initial capital but higher cash-to-asset ratios, because they expect faster future growth and want to avoid costly future equity issuance.
- This figure is the paper's core economic mechanism in one picture.

5.5 Table 3: baseline SMM fit

Read:

- The model matches many baseline moments well, including average cash at entry, cash autocorrelation, average equity issuance size, and mean sales growth at entry.
- The two main misses are also informative: the model makes firms deplete cash too quickly and fails to reproduce the small positive autocorrelation in sales growth.
- Baseline parameter estimates are economically plausible: $\sigma = 0.156$, $\rho = 0.649$, $\alpha = 0.694$, $\eta = 0.092$, $\log \underline{q} = -0.774$, $\nu = 0.851$, and $f_e = 0.093$.

5.6 Tables 4 and 5 plus Figure 7: industry-cohort estimation and model fit by sector

Read:

- Figure 7 shows that the model qualitatively matches the upward trend in cash at entry and the upward trend in sales-growth volatility, especially for the R&D-intensive sector.

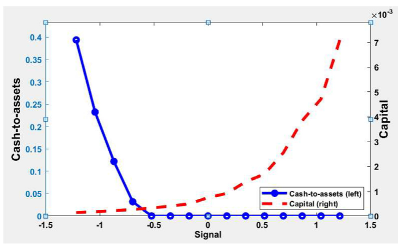

Fig. 6. Entry margin This figure reports the optimal cash-to-assets ratio (solid blue lien, left) and capital (dashed red line, right) at entry as a function of the (log) quality signal. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Simulated moments estimation: baseline cohort.

Panel A: Moments

Moments	Data	Model (N=532)	$t-stat$
Average cash holdings at entry	0.118	0.110	1.486
Average change in cash holdings	−0.018	−0.061	7.264
Volatility cash holdings	0.057	0.053	2.093
Autocorrelation cash holdings	0.276	0.282	−0.368
Average issue size at year 10	0.022	0.022	−0.067
Mean sales growth at 2	0.152	0.167	−1.152
Volatility sales growth at entry	0.086	0.054	3.426
Average change in sales growth	−0.136	−0.141	0.292
Volatility sales growth	0.191	0.175	2.941
Autocorrelation sales growth	0.053	−0.002	3.530
Average investment rate at entry	0.108	0.082	4.003
Average change in investment rate	−0.050	−0.056	0.746
Volatility investment rate	0.088	0.104	−5.972

Panel B: Parameter estimates

Parameter	Point estimate	S.E.
σ	0.156	0.002
ρ	0.649	0.009
α	0.694	0.011
η	0.092	0.009
ξ	11.804	14.640
$\log(\underline{q})$	−0.774	0.156
ν	0.851	0.034
f_e	0.093	0.057

This table reports the simulated and actual moments (Panel A) together with the estimated structural parameters (Panel B) using a balanced panel of firms belonging to the 1974-1978 cohort. The balanced panel is composed of all the firms that entered our sample of publicly listed firms during the period 1974-1978 and have ten consecutive years of observations on total assets (Compustat item *at*); revenues (item *sale*); cash (item *che*); net property, plant, and equipment (item *ppent*); common shares outstanding (item *csho*); sale of common and preferred stock (item *sstk*); and market price at the calendar year end (item *prcc_c*). In Panel A, we also report the number of firms in the sample (N) and the t-statistics for the difference between model-generated and actual moments. The estimated parameters in Panel B are the following: the volatility (σ) and persistence (ρ) of the TFPR process; the returns-to-scale parameter α; the convex adjustment cost parameter η; the shape parameter (ξ) and the lower bound ($\underline{q}$) of the Pareto distribution; the internal accumulation wedge ν; and the fixed equity issuance cost and f_e. For each estimated parameter, we report the associated standard error, whose calculation is described in Online Appendix C.2.

Table 4
Simulated moments estimation: industry 0.

Moments	1979-1983			1984-1988			1989-1993			1994-1998			1999-2003		
	Data	Model N=116	t-stat	Data	Model N=205	t-stat	Data	Model N=197	t-stat	Data	Model N=236	t-stat	Data	Model N=86	t-stat
Average cash holdings at entry	0.194	0.185	(0.50)	0.154	0.143	(0.96)	0.152	0.146	(0.48)	0.171	0.146	(2.06)	0.136	0.098	(1.99)
Average change in cash holdings	−0.083	−0.132	(2.82)	−0.054	−0.097	(3.72)	−0.073	−0.108	(2.65)	−0.049	−0.130	(6.08)	0.014	−0.077	(4.80)
Volatility cash holdings	0.085	0.084	(0.15)	0.070	0.067	(0.92)	0.070	0.070	(−0.12)	0.078	0.070	(2.23)	0.074	0.057	(2.58)
Autocorrelation cash holdings	0.287	0.376	(−2.63)	0.281	0.307	(−1.01)	0.267	0.284	(−0.66)	0.310	0.403	(−4.18)	0.270	0.252	(0.46)
Average issue size at year 10	0.021	0.021	(0.07)	0.020	0.020	(0.09)	0.019	0.019	(0.04)	0.015	0.015	(−0.05)	0.019	0.019	(0.03)
Mean sales growth at 2	0.186	0.199	(−0.36)	0.198	0.181	(0.71)	0.232	0.201	(1.29)	0.287	0.263	(1.06)	0.166	0.209	(−1.29)
Volatility sales growth at entry	0.142	0.098	(2.097)	0.113	0.068	(2.997)	0.111	0.091	(1.942)	0.124	0.115	(0.683)	0.093	0.062	(1.466)
Average change in sales growth	−0.127	−0.142	(0.33)	−0.126	−0.119	(−0.25)	−0.183	−0.167	(−0.51)	−0.222	−0.212	(−0.37)	−0.178	−0.198	(0.45)
Volatility sales growth	0.259	0.254	(0.30)	0.195	0.195	(−0.03)	0.227	0.236	(−0.93)	0.237	0.233	(0.38)	0.213	0.163	(2.80)
Autocorrelation sales growth	0.100	0.052	(1.46)	0.139	0.017	(5.04)	0.107	0.029	(2.94)	0.160	0.012	(6.55)	0.101	−0.026	(3.22)
Average investment rate at entry	0.157	0.147	(0.46)	0.116	0.102	(1.10)	0.149	0.124	(1.54)	0.142	0.140	(0.12)	0.132	0.119	(0.50)
Average change in investment rate	−0.106	−0.083	(−0.94)	−0.044	−0.040	(−0.24)	−0.080	−0.083	(0.17)	−0.094	−0.086	(−0.46)	−0.115	−0.104	(−0.43)
Volatility investment rate	0.140	0.174	(−3.43)	0.113	0.124	(−1.67)	0.130	0.146	(−2.04)	0.110	0.130	(−2.98)	0.088	0.092	(−0.41)

	σ	ρ	α	η	$\log(\underline{q})$	ν	f_e
1979-1983	0.246	0.795	0.493	0.051	−1.087	0.869	0.246
	0.007	0.017	0.005	0.000	0.144	0.018	0.114
1984-1988	0.186	0.774	0.606	0.072	−1.015	0.767	0.044
	0.003	0.009	0.017	0.007	0.027	0.007	0.042
1989-1993	0.221	0.721	0.592	0.044	−0.865	0.732	0.048
	0.005	0.014	0.007	0.006	0.039	0.052	0.015
1994-1998	0.215	0.717	0.564	0.071	−1.224	0.701	0.098
	0.002	0.013	0.000	0.032	0.198	0.032	0.060
1999-2003	0.125	0.498	0.715	0.056	−0.873	0.776	0.248
	0.005	0.016	0.024	0.002	0.161	0.160	1.785

This table reports the simulated and actual moments (Panel A) together with the estimated structural parameters (Panel B) using a balanced panel of firms in the non-R&D-intensive sector belonging to the following five cohorts: 1979-1983, 1984-1988, 1989-1993, 1994-1998, and 1999-2003. For each cohort, the balanced panel is composed of non-R&D-intensive firms that entered our sample of publicly listed firms and have ten consecutive years of observations on total assets (Compustat item *at*); revenues (item *sale*); cash (item *che*); net property, plant, and equipment (item *ppent*); common shares outstanding (item *csho*); sale of common and preferred stock (item *sstk*); and market price at the calendar year end (item *prcc_c*). In Panel A, we also report the number of firms in each cohort (N) and the t-statistics for the difference between model-generated and actual moments. The estimated cohort-level parameters in Panel B are the following: the volatility (σ) and persistence (ρ) of the TFPR process; the returns to scale parameter α; the convex adjustment cost parameter η; the shape parameter, (ξ) and the lower bound ($\underline{q}$) of the Pareto distribution; the internal accumulation wedge ν; and the fixed equity issuance cost and f_e. The parameters are separately estimated for each cohort. For each estimated parameter, we report the associated standard error, whose calculation is described in Online Appendix C.2.

Table 5
Simulated moments estimation: industry 1.

Moments	1979-1983			1984-1988			1989-1993			1994-1998			1999-2003		
	Data	Model N=172	t-stat	Data	Model N=227	t-stat	Data	Model N=213	t-stat	Data	Model N=306	t-stat	Data	Model N=199	t-stat
Average cash holdings at entry	0.283	0.298	(−0.75)	0.368	0.313	(2.98)	0.412	0.348	(3.14)	0.426	0.322	(6.30)	0.526	0.428	(4.67)
Average change in cash holdings	−0.093	−0.201	(5.22)	−0.114	−0.210	(6.27)	−0.110	−0.224	(5.92)	−0.097	−0.230	(9.09)	−0.172	−0.264	(4.42)
Volatility cash holdings	0.122	0.094	(5.26)	0.120	0.099	(4.74)	0.131	0.105	(5.30)	0.132	0.103	(7.64)	0.142	0.127	(3.00)
Autocorrelation cash holdings	0.333	0.380	(−1.70)	0.333	0.467	(−5.61)	0.337	0.455	(−4.89)	0.327	0.500	(−8.25)	0.322	0.552	(−9.38)
Average issue size at year 10	0.015	0.015	(0.07)	0.069	0.069	(0.04)	0.061	0.061	(−0.02)	0.063	0.062	(0.09)	0.055	0.055	(−0.03)
Mean sales growth at 2	0.263	0.254	(0.27)	0.319	0.328	(0.38)	0.279	0.293	(−0.47)	0.297	0.309	(−0.46)	0.346	0.325	(0.51)
Volatility sales growth at entry	0.187	0.109	(2.233)	0.206	0.164	(2.228)	0.181	0.164	(0.955)	0.209	0.155	(3.152)	0.329	0.188	(5.469)
Average change in sales growth	−0.195	−0.227	(0.88)	−0.205	−0.279	(1.99)	−0.243	−0.261	(0.47)	−0.186	−0.271	(2.59)	−0.325	−0.289	(−0.84)
Volatility sales growth	0.270	0.24	(2.35)	0.286	0.268	(1.34)	0.323	0.308	(1.11)	0.336	0.274	(5.14)	0.321	0.315	(0.48)
Autocorrelation sales growth	0.061	−0.047	(4.32)	0.095	−0.071	(6.68)	0.084	−0.156	(9.05)	0.077	−0.094	(8.59)	0.07	−0.089	(6.85)
Average investment rate at entry	0.117	0.114	(0.27)	0.081	0.081	(−0.02)	0.068	0.062	(0.76)	0.078	0.085	(−1.02)	0.048	0.057	(−1.30)
Average change in investment rate	−0.099	−0.089	(−0.84)	−0.045	−0.037	(−1.00)	−0.047	−0.033	(−1.41)	−0.069	−0.051	(−2.55)	−0.044	−0.022	(−2.96)
Volatility investment rate	0.093	0.111	(−4.20)	0.073	0.093	(−6.07)	0.07	0.089	(−5.36)	0.061	0.092	(−11.41)	0.048	0.097	(−15.22)

	σ	ρ	α	η	$\log(\underline{q})$	ν	f_e
1979-1983	0.209	0.513	0.823	0.062	−0.830	0.876	0.032
	0.005	0.011	0.013	0.002	0.023	0.013	0.326
1984-1988	0.229	0.511	0.850	0.179	−1.340	0.862	0.039
	0.002	0.012	0.012	0.005	0.089	0.012	0.037
1989-1993	0.263	0.461	0.888	0.165	−1.219	0.892	0.000
	0.008	0.012	0.002	0.004	0.059	0.002	0.001
1994-1998	0.236	0.494	0.851	0.159	−1.216	0.869	0.026
	0.005	0.008	0.001	0.071	0.050	0.035	0.005
1999-2003	0.275	0.505	0.872	0.192	−1.361	0.900	0.001
	0.005	0.008	0.005	0.022	0.092	0.101	0.001

This table reports the simulated and actual moments (Panel A) together with the estimated structural parameters (Panel B) using a balanced panel of firms in the R&D-intensive sector belonging to the following five cohorts: 1979-1983, 1984-1988, 1989-1993, 1994-1998, and 1999-2003. For each cohort, the balanced panel is composed of R&D-intensive firms that entered our sample of publicly listed firms and have ten consecutive years of observations on total assets (Compustat item *at*); revenues (item *sale*); cash (item *che*); net property, plant, and equipment (item *ppent*); common shares outstanding (item *csho*); sale of common and preferred stock (item *sstk*); and market price at the calendar year end (item *prcc_c*). In Panel A, we also report the number of firms in each cohort (N) and the t-statistics for the difference between model-generated and actual moments. The estimated cohort-level parameters in Panel B are the following: the volatility (σ) and persistence (ρ) of the TFPR process; the returns to scale parameter α; the convex adjustment cost parameter η; the shape parameter (ξ) and the lower bound ($\underline{q}$) of the Pareto distribution; the internal accumulation wedge ν; and the fixed equity issuance cost and f_e. The parameters are separately estimated for each cohort. For each estimated parameter, we report the associated standard error, whose calculation is described in Online Appendix C.2.

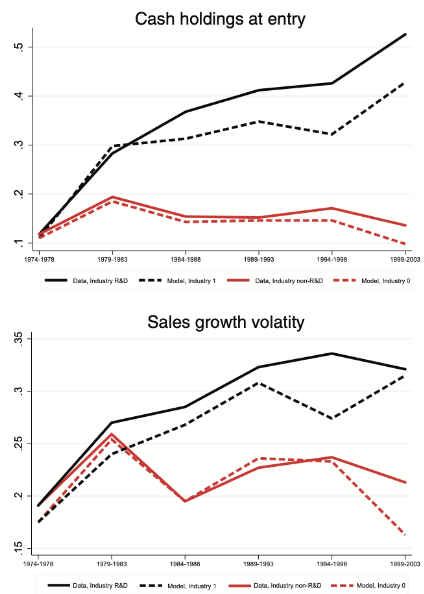

Fig. 7. Secular trends: model versus data. This figure reports model-generated average cash-to-asset ratio at entry (top panel) and average firm-level sales growth volatility (bottom panel) and compares these quantities with their empirical counterparts. The quantities are reported for each cohort, starting from the 1974-1978 one, and across two industries. The black lines refer to industry 1 in the model and R&D-intensive industry in the data, while the red lines refer to industry 0 in the model and non-R&D-intensive industry in the data. Solid lines refer to actual data, while dashed lines refer to model-generated data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

- Table 4 shows that non-R&D cohorts change relatively little over time. Table 5 shows that R&D-intensive cohorts exhibit progressively higher entry cash and much higher sales-growth volatility.
- Relative to industry 0, industry 1 firms are estimated to be smaller, more volatile, less persistent, and closer to constant returns: roughly $\sigma \approx 0.24$ versus 0.20, $\rho \approx 0.50$ versus 0.70, and $\alpha \approx 0.86$ versus 0.59.
- The estimated lower bound of the entrant productivity signal is also much lower for industry 1, especially in later cohorts. That is the direct structural counterpart to the descriptive fact that later R&D entrants enter public markets with higher cash and higher growth potential.

5.7 Table 6 and Table 7: aggregate simulation over 1979–2003

Read:

- Table 6 feeds the observed changing composition of entrants into the simulation: the R&D share of entrants rises from 0.33 in 1974–1978 to 0.74 in 1999–2003.
- Table 7 shows that the simulated economy reproduces a large fraction of the aggregate secular trends.
- In the data, average cash rises by 0.136 (a 160% increase). In the model, it rises by 0.056 (a 78% increase).
- In the data, sales-growth volatility rises by 0.152 (an 88% increase). In the model, it rises by 0.078 (a 44% increase).
- The main shortfall is that the model does not generate a big enough increase in cash for later R&D-intensive cohorts and depletes incumbents' cash too quickly.

Table 6
Entrants distribution.

Cohort	1974-1978	1979-1983	1984-1988	1989-1993	1994-1998	1999-2003
Industry 1 (ω)	0.33	0.55	0.45	0.52	0.55	0.74
Industry 0 (1-ω)	0.67	0.45	0.55	0.48	0.45	0.26

This table reports the composition on firms entering in U.S. public equity markets during the period 1974-2003 across six cohorts of entrants: 1974-1978, 1979-1983, 1984-1988, 1989-1993, 1994-1998, and 1999-2003. For each cohort, we report the fraction of firms belonging to the R&D-intensive industry (ω) and the fraction of firms belonging to the non-R&D-intensive industry ($1-\omega$).

Table 7
Cash and volatility dynamics.

Year5	Sample composition		Cash-to-asset		Cash-to-asset (Ind. 0)		Cash-to-asset (Ind. 1)		Sales growth vol	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Data	Model	Data	Model	Data	Model	Data	Model	Data	Model
1974-1978	0.336	0.330	0.085	0.071	0.085	0.071	0.085	0.071	0.172	0.176
1979-1983	0.376	0.381	0.108	0.090	0.095	0.066	0.126	0.102	0.223	0.192
1984-1988	0.434	0.417	0.140	0.096	0.108	0.068	0.182	0.110	0.286	0.215
1989-1993	0.467	0.443	0.153	0.104	0.101	0.068	0.210	0.130	0.269	0.228
1994-1998	0.500	0.476	0.182	0.105	0.092	0.060	0.270	0.135	0.306	0.244
1999-2003	0.554	0.543	0.221	0.127	0.097	0.044	0.322	0.187	0.324	0.254
Difference	0.218	0.213	0.136	0.056	0.012	−0.027	0.237	0.116	0.152	0.078
% change	65%	65%	160%	78%	14%	−38%	278%	163%	88%	44%

This table reports simulated quantities across 50 simulations. For each simulation, we simulate a panel with 4,000 firms. In each period, a firm can exit the sample exogenously with a 8% probability and is replaced by a new entrant, which chooses initial capital and cash balances according to Eq. 5. We first simulate the economy using the estimated parameters for the baseline cohort for 350 periods. Then, we simulate a period of length 25 years and let the firms of different industries and cohorts enter over time with composition at entry dictated by the quantities in Table 6. We report the model-generated quantities together with their empirical counterparts. All quantities are calculated at an annual frequency and are averaged over a five-year period. The first two columns report the fraction of incumbents firms belonging to the R&D-intensive industry in the data (Column 1) and to industry 1 in model (Column 2). Columns 3 and 4 report the cash-to-asset ratio's cross-sectional average. Columns 5 and 6 report the cash-to-asset ratio's cross-sectional average for firms belonging to the non-R&D-intensive industry in the data (Column 5) and to industry 0 in model (Column 6). Columns 7 and 8 report the cash-to-asset ratio's cross-sectional average for firms belonging to the R&D-intensive industry in the data (Column 7) and to industry 0 in model (Column 8). Columns 9 and 10 report the average firm-level sales growth rate volatility. All model-based quantities are averaged over the 50 simulations.

5.8 Table 8: the role of selection

Read:

- If the lower bound of entrant productivity $\underline{q}$ is fixed at its baseline value, the model still generates only 0.023 extra cash instead of 0.056. So changes in entrant quality are doing a lot of work.
- If both entrant composition (ω) and entrant productivity ($\underline{q}$) are fixed at baseline values, the model generates no secular increase in cash at all; in fact the change becomes slightly negative.
- If only entrant composition and entrant productivity are allowed to move, the model generates a cash increase of 0.035, or about 50% in percentage terms, which is roughly 60% of the full model response.
- Selection alone also explains part of the rise in sales-growth volatility, but much less than it explains the rise in cash.

Table 8
The role of selection.

	Cash-to-asset				Sales growth dispersion			
	Model (1)	$\underline{q}$ fixed (2)	(ω, $\underline{q}$) fixed (3)	only (ω, $\underline{q}$) (4)	Model (5)	$\underline{q}$ fixed (6)	(ω, $\underline{q}$) fixed (7)	only (ω, $\underline{q}$) (8)
1974-1978	0.071	0.071	0.071	0.071	0.176	0.176	0.176	0.176
1979-1983	0.090	0.095	0.080	0.082	0.192	0.196	0.193	0.181
1984-1988	0.096	0.076	0.069	0.097	0.215	0.210	0.208	0.189
1989-1993	0.104	0.079	0.070	0.094	0.228	0.221	0.218	0.189
1994-1998	0.105	0.077	0.064	0.109	0.244	0.236	0.230	0.193
1999-2003	0.127	0.094	0.064	0.107	0.254	0.246	0.231	0.199
Difference	0.056	0.023	−0.008	0.035	0.078	0.070	0.055	0.022
% change	78%	32%	−11%	50%	44%	40%	31%	13%

This table reports simulated quantities across 50 simulations. For each simulation, we simulate a panel with 4,000 firms. In each period, a firm can exit the sample exogenously with a 8% probability and is replaced by a new entrant, which chooses initial capital and cash balances according to Eq. 5. We first simulate the economy using the estimated parameters for the baseline cohort for 350 periods. Then, we simulate a period of length 25 years and let the firms of different industries and cohorts enter over time with composition at entry dictated by the quantities in Table 6. The first four columns report the cash-to-asset ratio's cross-sectional average, while the last four columns report the average firm-level sales growth rate volatility. Columns 1 and 5 report quantities calculated using the parameters estimated using SMM. Columns 2 and 6 report quantities from a model in which all parameters are kept at their estimated values except for $\underline{q}$. Columns 3 and 7 report quantities from a model in which all parameters are kept at their estimated values except for $\underline{q}$ and the fraction of entrants from industry 1, kept at its baseline value (0.33). Columns 4 and 8 report quantities from a model in which all parameters are set to their baseline value except for the composition at entry and $\underline{q}$. All are averaged over the 50 simulations.

Table 9
The role of cash flow volatility.

Panel A: Cash flow volatility

	Cash-to-asset			Sales growth dispersion		
	Model (1)	σ fixed (2)	only σ (3)	Model (4)	σ fixed (5)	only σ (6)
1974-1978	0.071	0.071	0.071	0.176	0.176	0.176
1979-1983	0.090	0.082	0.085	0.192	0.184	0.191
1984-1988	0.096	0.087	0.083	0.215	0.190	0.208
1989-1993	0.104	0.088	0.088	0.228	0.190	0.218
1994-1998	0.105	0.085	0.094	0.244	0.192	0.229
1999-2003	0.127	0.113	0.088	0.254	0.197	0.229
Difference	0.056	0.042	0.017	0.078	0.021	0.053
% change	78%	59%	24%	44%	12%	30%

Panel B: Selection and cash flow volatility

	Cash-to-asset			Sales growth dispersion		
	Model (1)	(σ, ω, $\underline{q}$) fixed (2)	only (σ, ω, $\underline{q}$) (3)	Model (4)	(σ, ω, $\underline{q}$) fixed (5)	only (σ, ω, $\underline{q}$) (6)
1974-1978	0.071	0.071	0.071	0.176	0.176	0.176
1979-1983	0.090	0.064	0.097	0.192	0.180	0.195
1984-1988	0.096	0.049	0.123	0.215	0.177	0.221
1989-1993	0.104	0.048	0.125	0.228	0.178	0.233
1994-1998	0.105	0.040	0.139	0.244	0.178	0.247
1999-2003	0.127	0.040	0.144	0.254	0.180	0.256
Difference	0.056	−0.031	0.072	0.078	0.004	0.080
% change	78%	−44%	102%	44%	2%	45%

This table reports simulated quantities across 50 simulations. For each simulation, we simulate a panel with 4,000 firms. In each period, a firm can exit the sample exogenously with a 8% probability and is replaced by a new entrant, which chooses initial capital and cash balances according to Eq. 5. We first simulate the economy using the estimated parameters for the baseline cohort for 350 periods. Then, we simulate a period of length 25 years and let the firms of different industries and cohorts enter over time with composition at entry dictated by the quantities in Table 6. The first three columns report the cash-to-asset ratio's cross-sectional average, while the last three columns report the average firm-level sales growth rate volatility. Columns 1 and 4 in both Panel A and Panel B report quantities calculated using the parameters estimated using SMM. In Panel A, Columns 2 and 5 report quantities from a model in which all parameters are kept at their estimated values except for σ, while Columns 3 and 6 report quantities from a model in which all parameters are set to their baseline value except for σ. In Panel B, Columns 2 and 5 report quantities from a model in which all parameters are kept at their estimated values except for σ, $\underline{q}$, and the fraction of entrants from industry 1, kept at its baseline value (0.33). Columns 3 and 6 in Panel B report quantities from a model in which all parameters are set to their baseline value except for σ, $\underline{q}$, and composition at entry. All are averaged over the 50 simulations.

5.9 Table 9: the role of cash-flow volatility

Read:

- If the volatility parameter σ is fixed at its baseline value while other estimated forces remain active, the model still produces a sizable rise in cash (0.042) but almost no rise in sales-growth volatility (0.021).

- If only σ is allowed to vary over time, the model produces only a small rise in cash (0.017) but a substantial rise in sales-growth volatility (0.053).
- When entrant composition, entrant productivity, and cash-flow volatility are all allowed to move jointly, the model produces a cash increase of 0.072 and a volatility increase of 0.080.
- So selection is the main force behind higher cash holdings, while higher cash-flow volatility is the main force behind higher sales-growth volatility.

6 Short summary

- **Conclusion.** The secular rise in cash holdings among U.S. public firms is primarily a selection story: later entrants, especially R&D-intensive entrants, come public with much higher cash balances.
- **Structural insight.** A simple dynamic corporate finance model with costly equity issuance, costly cash carry, decreasing returns, mean reversion, and heterogeneous entry can replicate this pattern naturally.
- **Quantitative lesson.** Selection explains most of the model-generated rise in cash, while rising cash-flow volatility is essential for explaining the parallel rise in sales-growth volatility.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that the long-run rise in cash among U.S. public firms cannot be understood primarily as incumbent firms hoarding more cash over time.
- Instead, the main force is a change in the type of firms entering public markets: progressively smaller, riskier, and less initially productive firms—especially in R&D-intensive industries—enter with larger cash buffers.
- A standard neoclassical investment model augmented with heterogeneous entry is capable of explaining roughly half of the observed secular increase in cash holdings and a substantial part of the rise in sales-growth volatility.

7.2 Economic intuition

- A low-productivity entrant is far below its long-run scale. Because productivity mean reverts, that entrant expects strong future growth.
- Strong future growth means strong future financing needs. If external equity is costly, the entrant prefers to go public holding extra cash today.
- Once the firm grows and becomes better able to fund itself internally, its precautionary cash motive weakens, so the cash-to-asset ratio falls. This delivers the negative within-firm trend.
- When public markets tilt toward these sorts of entrants, the cross-sectional average cash ratio rises even if each individual entrant later runs cash down.

7.3 Takeaway

This paper is best read as a structural explanation for why average public-firm cash can rise while individual firms' cash ratios fall. Its central message is that sample composition matters enormously in corporate finance. The public-firm universe after 1980 is not just an older universe with more cash; it is a different universe, populated by more R&D-intensive, smaller, riskier, and higher-growth firms. The selection mechanism is therefore not a side issue—it is a first-order driver of the secular cash-holdings fact.